Computing Master Feynman Integrals with Positive Semi-Definite Constraints

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Background

This work is based on [Zeng(2023)].

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Feynman Integrals from Positivity Constraints

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Abstract: We explore inequality constraints as a new tool for numerically evaluating Feynman integrals. A convergent Feynman integral is non-negative if the integrand is non-negative in either loop momentum space or Feynman parameter space. Applying various identities, all such integrals can be reduced to linear sums of a small set of master integrals, leading to infinitely many linear constraints on the values of the master integrals. The constraints can be solved as a semidefinite programming problem in mathematical optimization, producing rigorous two-sided bounds for the integrals which are observed to converge rapidly as more constraints are included, enabling high-precision determination of the integrals. Positivity constraints can also be formulated for the ϵ expansion terms in dimensional regularization and reveal hidden consistency relations between terms at different orders in ϵ. We introduce the main methods using one-loop bubble integrals, then present a nontrivial example of three-loop banana integrals with unequal masses, where 11 top-level master integrals are evaluated to high precision.

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Feynman integrals from positivity constraints

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Evaluating Feynman integrals

- A classic problem, yet still yielding active progress with **computers**.
- Representative algorithms:

  Direct evaluation: Feynman parametrization, sector decomposition, Mellin-Barnes representation, ...

  Evaluation via reduction: IBP reduction, diffential equations, dimensional recurrence, ...

Semi-definite constraints

- A method proposed in [Zeng(2023)].
- Leveraging **inequality constraints** on Feynman integrals.

- A toy example:

$$
\begin{aligned}
I^d_{a_1,a_2} &= \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{(-\ell^2+m^2)^{a_1}[-(p+\ell)^2+m^2]^{a_2}} \\
&= \frac{\Gamma(a_1+a_2-d/2)}{\Gamma(a_1)\Gamma(a_2)} \int_0^1 \frac{x^{a_1-1}(1-x)^{a_2-1}}{\mathcal{F}(x)^{a_1+a_2-d/2}} dx,
\end{aligned}
$$

where $\mathcal{F}(x) = m^2 - p^2 x(1-x)$.

Semi-definite constraints

Consider kinematics $p^2 < 4m^2$. As a result, $\mathcal{F}(x) \geqslant 0$ for any $0 \leqslant x \leqslant 1$. We have

$$\int_0^1 \left(c_1 + c_2 \frac{x}{\mathcal{F}(x)} + c_3 \frac{1-x}{\mathcal{F}(x)}\right)^2 \frac{x^{a_1-1}(1-x)^{a_2-1}}{\mathcal{F}(x)^{a_1+a_2-d/2}} dx \geqslant 0,$$

or equivalently, denoting $\hat{I}^d_{a_1,a_2} = \frac{\Gamma(a_1)\Gamma(a_2)}{\Gamma(a_1+a_2-d/2)} I^d_{a_1,a_2}$,

$$[c_1, c_2, c_3] \begin{bmatrix} \hat{I}^d_{a_1,a_2} & \hat{I}^d_{a_1+1,a_2} & \hat{I}^d_{a_1,a_2+1} \\ \hat{I}^d_{a_1+1,a_2} & \hat{I}^d_{a_1+2,a_2} & \hat{I}^d_{a_1+1,a_2+1} \\ \hat{I}^d_{a_1,a_2+1} & \hat{I}^d_{a_1+1,a_2+1} & \hat{I}^d_{a_1,a_2+2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \geqslant 0.$$

Semi-definite constraints

- Every entry of the LHS matrix can be reduced to master integrals.
- LHS gives **semi-definite constraints** on master integrals.

Two constraints:

- **Euclidean kinematics:** $\mathcal{F}(x) \geqslant 0$ throughout integral range of Feynman parameter x.
- **Convergence:** All integrals appearing at LHS (including all master integrals) converge.

Our contribution

Problem with [Zeng(2023)]:

- **No public implementation.** Code released with [Zeng(2023)] only treats a few examples and does not work generally.
- **Limited applicability.** The method cannot deal with Feynman integrals with **divergences**, which are ubiquitous.

Our work:

- Gives the first general `C++` implementation for [Zeng(2023)].
- Generalize the method to integrals with arbitrary divergences.

Workflow

Three steps to evaluate a set of master integrals.

Step 1:
Integral reduction

Step 2:
Run our program

config/jobs.yaml ⇒ **Kira** ⇒ kira_bubble.m &

bubble.yaml

```
integralfamily:
  name: bubble
  internals: [l]
  externals: [p]
  kinematic_invariants:
    - [s, 2]
    - [m, 1]
  propagators:
    - [l, m^2]
    - [l+p, m^2]
  scalarproduct_rules:
    - [[p, p], s]
kiradir: /root/bubble
kirafile: kira_bubble.m
kinematics_numerics:
  - [s, 2]
  - [m, 1]
d0: 4
t: 3
eps_order: 2
master_values:
  - [[3, 0], 1]
ansatze:
  - [x0, 0, 5, 1]
  - [x1, 0, 5, 1]
```

Workflow

Three steps to evaluate a set of master integrals.

Step 2:
Run our program

kira_
bubble.m &

bubble.yaml


```yaml
integralfamily:
  name: bubble
  internals: [l]
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  kinematic_invariants:
    - [s, 2]
    - [m, 1]
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    - [l, m^2]
    - [l+p, m^2]
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    - [[p, p], s]
kiradir: /root/bubble
kirafile: kira_bubble.m
kinematics_numerics:
  - [s, 2]
  - [m, 1]
d0: 4
t: 3
eps_order: 2
master_values:
  - [[3, 0], 1]
ansatze:
  - [x0, 0, 5, 1]
  - [x1, 0, 5, 1]
```


Step 3:
Run SDPA solver

$\Rightarrow$ problem.in
param.sdpa

Workflow

Three steps to evaluate a set of master integrals.

Step 3:
Run SDPA[Yamashita et al.(2012)] solver

`problem.in`
`param.sdpa`
	⇒	`sdpa`
		`sdpa_dd`
		`sdpa_qd`
			⇒	`problem.out`
				(master integral results)

General treatment

What does our program do?

- Feynman parametrize every relevant integral.

$$
\begin{aligned}
\hat{I}^{d}_{a_1,\ldots,a_n} &= \frac{\prod_k \Gamma(a_k)}{\Gamma(a - Ld/2)} \left(\prod_j \int \frac{d^d \ell_j}{i\pi^{d/2}} \right) \frac{1}{\prod_k D_k^{a_k}} \\
&= \int_{\boldsymbol{x} \geqslant 0} d^n \boldsymbol{x} \; \delta\left(1 - \sum_k x_k\right) \prod_k \tilde{x}_k^{a_k - 1} \, \frac{\mathcal{U}(\boldsymbol{x})^{n-(L+1)d/2}}{\mathcal{F}(\boldsymbol{x})^{n-Ld/2}}
\end{aligned}
$$

assuming $a_k \geqslant 1$ for $k = 1, \ldots, n$. Here $\tilde{x}_k \equiv (\mathcal{U}(\boldsymbol{x})/\mathcal{F}(\boldsymbol{x}))x_k$.

General treatment

What does our program do?

- Derive semi-definite constraints.

$$0 \leqslant \int_{\boldsymbol{x}\geqslant 0} d^n \boldsymbol{x}\ \delta\left(1-\sum_k x_k\right) \prod_k \tilde{x}_k^{a_k-1}\ P^2(\tilde{\boldsymbol{x}})\ \frac{\mathcal{U}(\boldsymbol{x})^{n-(L+1)d/2}}{\mathcal{F}(\boldsymbol{x})^{n-Ld/2}}.$$

Conditions:

- **Euclidean kinematics:** $\mathcal{F}(\boldsymbol{x}) \geqslant 0$ over all possible $\boldsymbol{x}$.
- **Convergence:** all relevant integrals at RHS converge.

General treatment

What does our program do?

- Reduce all integrals at RHS, and solve the resulting semi-definite programming problem.

Remarks:

- The method can compute not only $\hat{I}^{d_0}_{a_1,\ldots,a_n}$, but also $\frac{\partial^t}{\partial \epsilon^t}\hat{I}^{d}_{a_1,\ldots,a_n}\Big|_{\epsilon=0}$ where $d = d_0 - 2\epsilon$ (i.e., ϵ-expansion).
- $P(\tilde{\boldsymbol{x}})$ is taken as a linear combination of terms (such as $c_1 + c_2\tilde{x}_1 + c_3\tilde{x}_1\tilde{x}_2 + \cdots$).

  Example: `ansatze: [x0, 0, 5, 0]` means

$$\prod_k \tilde{x}_k^{a_k-1}\, P^2(\tilde{\boldsymbol{x}}) = \tilde{x}_1 \left(\sum_{0\leqslant b_1+\cdots+b_n\leqslant 5} c_{b_1,\ldots,b_n} \prod_k \tilde{x}_k^{b_k} \right)^2$$

Treating divergent integrals

Feynman integrals usually have **divergences**.

- **UV divergences:** divergences appearing when some loop momenta go ∞, and **not** related to kinematic singularities.
- **IR divergences:** divergences related to kinematic singularities.

We generalize [Zeng(2023)] to divergent master integrals!

UV divergent integrals

Master integrals with **only** UV divergences (e.g., massive integrals) are easy to tackle!

- **Step 1:** lower dimension $d \to d - 2r$ (r is a positive integer) to make master integrals UV finite.
- **Step 2:** evaluate at dimension $d - 2r$.
- **Step 3:** use **dimensional-shifting relations** to move from dimension $d - 2r$ to d.

UV divergent integrals – Example

Consider at $d = 4 - 2\epsilon$,

$$J^d_{a_1,a_2,a_3,a_4} = \left(\prod_{j=1}^{3} \int \frac{d^d \ell_j}{i\pi^{d/2}}\right) \frac{1}{(-\ell_1^2 + m_1^2)^{a_1}} \frac{1}{(-\ell_2^2 + m_2^2)^{a_2}} \times \frac{1}{(-\ell_3^2 + m_3^2)^{a_3}} \frac{1}{[-(p + \ell_1 + \ell_2 + \ell_3)^2 + m_4^2]^{a_4}}.$$

Master integrals are chosen as

$$J^d_{1,1,1,1}, \quad J^d_{2,2,2,0}, \quad J^d_{1,1,1,2}, \quad \ldots$$

which are UV divergent at $d = 4 - 2\epsilon$ but converge at $d = 2 - 2\epsilon$.

$\Longrightarrow$ compute at $d = 2 - 2\epsilon$, and then shift to $d = 4 - 2\epsilon$

IR divergent integrals

Master integrals with IR (kinematic) divergences are harder!

- Raising dimension does not always work (potentially introducing UV divergences).
- Semi-definite constraints still brings **boundary conditions**!

IR divergent integrals – Example

Consider at $d = 2 - 2\epsilon$,

$$J^d_{a_1,a_2,a_3,a_4} = \left(\prod_{j=1}^{3} \int \frac{d^d \ell_j}{i\pi^{d/2}}\right) \frac{1}{(-\ell_1^2)^{a_1}} \frac{1}{(-\ell_2^2 + m_2^2)^{a_2}} \times \frac{1}{(-\ell_3^2 + m_3^2)^{a_3}} \frac{1}{[-(p + \ell_1 + \ell_2 + \ell_3)^2 + m_4^2]^{a_4}}.$$

Master integrals are

$$J^d_{1,1,1,1}, \quad J^d_{0,2,2,2}, \quad J^d_{1,1,1,2}, \quad \ldots$$

which are UV finite but IR divergent (when $\ell_1^2 \to 0$, except $J^d_{0,2,2,2}$).

IR divergent integrals – Treatment

- **Step 1:** Introduce auxiliary mass $\eta > 0$.

$$\tilde{J}^d_{a_1,a_2,a_3,a_4}(\eta) = \left(\prod_{j=1}^{3}\int \frac{d^d\ell_j}{i\pi^{d/2}}\right) \frac{1}{(-\ell_1^2+\eta)^{a_1}} \frac{1}{(-\ell_2^2+m_2^2)^{a_2}} \times \frac{1}{(-\ell_3^2+m_3^2)^{a_3}} \frac{1}{[-(p+\ell_1+\ell_2+\ell_3)^2+m_4^2]^{a_4}}.$$

In our case, $\tilde{J}^d_{a_1,a_2,a_3,a_4}(\eta)$ is IR finite for every $\eta > 0$.

Note: This step usually increases the number of master integrals. (For example, $J^d_{2,2,2,0}$ is scaleless and vanishes, but $\tilde{J}^d_{2,2,2,0}(\eta)$ can be a master integral.)

IR divergent integrals – Treatment

- **Step 2:** Find master integrals $\tilde{\vec{J}}^d(\eta)$ of $\{\tilde{J}^d_{a_1,\ldots,a_n}(\eta)\}$ and build **differential equation** of $\tilde{\vec{J}}^d(\eta)$ with respect to η.

$$\frac{\partial}{\partial \eta}\tilde{\vec{J}}^d(\eta) = A(\eta;\epsilon)\tilde{\vec{J}}^d(\eta).$$

- **Step 3:** Find **general solution** for $\tilde{\vec{J}}^d(\eta)$ of form

$$\tilde{\vec{J}}^d(\eta) = \sum_{\alpha,\beta,\lambda} \eta^{\alpha+\beta\epsilon} \log^\lambda \eta \sum_{\mu=0}^{\infty} \vec{c}_{\alpha,\beta,\lambda,\mu}(\epsilon)\eta^\mu.$$

IR divergent integrals – Treatment

In our example, $\tilde{\vec{J}}^d(\eta)$ includes 15 master integrals, and the general solution has form

$$\tilde{\vec{J}}^d(\eta) = \sum_{i=1}^{7} f_i \left(\eta^{-1-\epsilon} \sum_{\mu=0}^{\infty} \vec{c}_{i,\mu} \eta^{\mu} \right) + \sum_{i=1}^{8} g_i \left(\sum_{\mu=0}^{\infty} \vec{d}_{i,\mu} \eta^{\mu} \right),$$

where $f_1, \ldots, f_7, g_1, \ldots, g_8$ are 15 arbitrary constants to be determined.

IR divergent integrals – Treatment

- **Step 4:** Choose $\eta_1 > 0$ within radius of convergence η_0, evaluate $\tilde{\vec{J}}^d(\eta_1)$ and determine arbitrary constants in the general solution.

- **Step 5:** Take limit $\eta \to 0^+$,

$$\lim_{\eta \to 0^+} \tilde{\vec{J}}^d(\eta) = \vec{c}_{0,0,0,0}(\epsilon).$$

This is our desired result for master integrals of $\{J^d_{a_1,\ldots,a_n}\}$.

In our example, only $g_1, \ldots, g_8$ are relevant to our concerned result.

Summary

Our current work

- implements a method to compute master integrals from semi-definite constraints;
- generalizes the method to arbitrarily divergent integrals.

Outlooks:

- Is it possible to remove Euclidean condition?
- Is it necessary reducing all integrals to master integrals (which can be very costly)?

Thank you for listening!

M. Yamashita et al.
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